

Document 06: System Architecture & Technical Design

Project: Natural Intellects School Management System (NISMS)

Version: 1.0

Status: Draft

Purpose:

This document defines the architectural principles, technical standards, deployment strategies, infrastructure considerations, scalability requirements, and implementation guidelines for the Natural Intellects School Management System.

This document serves as the primary reference for:

- Cloud AI implementation
- Database design
- Backend development
- Frontend development
- DevOps
- Security planning
- Future scalability

TABLE OF CONTENTS

1. Architectural Philosophy
2. Deployment Model
3. Multi-Tenant Strategy
4. Technology Stack
5. Frontend Architecture
6. Backend Architecture
7. Database Architecture
8. Authentication & Authorization
9. File Management

- 10. Website Architecture
 - 11. Mobile Responsiveness Strategy
 - 12. Security Principles
 - 13. Backup & Recovery
 - 14. Scalability Considerations
 - 15. Future Technical Expansion
-

1. ARCHITECTURAL PHILOSOPHY

The system shall prioritize:

Simplicity

Architecture must remain understandable and maintainable.

Avoid:

- Unnecessary microservices
 - Excessive complexity
 - Premature optimization
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Modularity

Major modules shall remain independent.

Examples:

- Students
- Teachers
- Attendance
- Fees
- Academics
- Announcements
- Website

Modules shall communicate through clearly defined boundaries.

Mobile-First Development

All interfaces shall work effectively on:

- Smartphones
 - Tablets
 - Laptops
 - Desktop computers
-

Historical Data Preservation

No major operational records shall be permanently deleted.

The system shall prefer:

- Archiving
 - Status changes
 - Immutable transaction histories
-

2. DEPLOYMENT MODEL

Recommended Approach

Natural Intellects shall initially support:

One Deployment Per School

Examples:

```
stmarysschool.naturalintellects.com
```

```
greenhillsschool.naturalintellects.com
```

Advantages:

- Easier maintenance
 - Better isolation
 - Simpler backups
 - Improved security
 - Easier customization
-

Alternative Future Model

Potential SaaS Model:

app.naturalintellects.com

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Multiple Schools

This shall be considered in future phases.

Strategic Decision

Version 1 shall prioritize:

ONE SCHOOL

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ONE DATABASE

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ONE DEPLOYMENT

3. MULTI-TENANT STRATEGY

Current Position

The first implementation shall avoid full multi-tenancy.

Reasons:

- Easier customer onboarding.
 - Simpler troubleshooting.
 - Reduced architectural complexity.
 - Better customer-specific customization.
-

Future Considerations

Possible future support:

- Shared infrastructure
- Tenant isolation
- White-label SaaS offerings

These shall remain outside current scope.

4. TECHNOLOGY STACK

Frontend

Recommended:

Next.js

Reasons:

- Modern ecosystem
 - Strong SEO support
 - Excellent website capabilities
 - Good Cloud AI compatibility
 - Strong TypeScript support
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Backend

Recommended:

Next.js API Routes

Reasons:

- Simpler deployment
 - Reduced infrastructure complexity
 - Shared codebase
-

Database

Recommended:

PostgreSQL

Reasons:

- Reliability
 - Strong relational capabilities
 - Scalability
 - Excellent support for educational systems
-

ORM

Recommended:

Prisma ORM

Reasons:

- Strong type safety
 - Better maintainability
 - AI-friendly schema generation
-

Authentication

Recommended:

Better Auth

Alternative:

Clerk

Requirements:

- Role-based access control
 - Session management
 - Password recovery
-

Storage

Recommended:

Supabase Storage

Use Cases:

- Student photos
- Teacher photos

- School logos
 - Gallery images
 - Documents
 - Report exports
-

Hosting

Recommended:

Frontend:

Vercel

Database:

Supabase PostgreSQL

Storage:

Supabase Storage

5. FRONTEND ARCHITECTURE

Public Website

Examples:

```
/
/about
/admissions
/academics
/news
/gallery
/contact
```

Admin Portal

Examples:

```
/admin  
  
/admin/students  
  
/admin/teachers  
  
/admin/fees  
  
/admin/reports
```

Teacher Portal

Examples:

```
/teacher  
  
/teacher/classes  
  
/teacher/attendance  
  
/teacher/marks
```

Navigation Principles

The system shall prioritize:

- Minimal clicks
 - Clear labels
 - Mobile accessibility
 - Fast access to common actions
-

6. BACKEND ARCHITECTURE

Modular Design

Examples:

Students Module

Teachers Module

Attendance Module

Fees Module

Reports Module

Announcements Module

API Organization

Example:

/api/students

/api/teachers

/api/attendance

/api/fees

/api/reports

Business Logic Separation

The system shall separate:

Controllers

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Services

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Database Operations

This improves maintainability and Cloud AI implementation quality.

7. DATABASE ARCHITECTURE

Core Entities

Examples:

Students
Teachers
Classes
Streams
Subjects
Attendance
Payments
Guardians
Announcements

Relationships

Examples:

One Student
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Many Payments

One Student
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Many Attendance Records

One Guardian

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Many Students

One Teacher

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Many Subjects

Database Principles

The database shall prioritize:

- Data integrity
- Historical preservation
- Strong relationships
- Minimal duplication

8. AUTHENTICATION & AUTHORIZATION

Authentication Methods

Support:

- Email login
- Password login

Future Possibilities:

- Phone number authentication
 - OTP verification
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User Roles

Examples:

- Super Admin
 - Proprietor
 - Head Teacher
 - Secretary
 - Bursar
 - Teacher
-

Authorization Rules

Permissions shall follow:

Role-Based Access Control (RBAC).

Users shall only access features relevant to their responsibilities.

9. FILE MANAGEMENT

Student Assets

Examples:

- Photos
 - Documents
 - Transfer letters
-

Teacher Assets

Examples:

- Photos
 - Employment documents
-

School Assets

Examples:

- Logos

- Gallery images
 - Admission forms
-

Storage Principles

The system shall:

- Separate files from database records.
 - Use cloud storage.
 - Support backups.
-

10. WEBSITE ARCHITECTURE

CMS Approach

Administrators shall manage:

- News
- Gallery
- Admissions information
- School contacts

without requiring technical expertise.

SEO Requirements

The website shall support:

- Search engine indexing
 - Mobile optimization
 - Fast loading speeds
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Branding Support

Schools shall customize:

- Logos
 - Colors
 - Contact information
-

11. MOBILE RESPONSIVENESS STRATEGY

Priority Devices

Examples:

- Android phones
 - Tablets
 - Laptops
 - Desktop computers
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Design Principles

Examples:

- Large buttons
 - Clear typography
 - Touch-friendly controls
 - Minimal scrolling
-

Performance Requirements

Pages shall load quickly under:

- Limited internet conditions
 - Low-end devices
-

12. SECURITY PRINCIPLES

Password Security

Requirements:

- Secure hashing
 - Strong password policies
 - Session expiration
-

Access Controls

Users shall only access:

Authorized modules.

Audit Trails

Future support may include:

- User activity logs
 - Financial logs
 - Administrative actions
-

Data Protection

The system shall prioritize:

- Student privacy
 - Financial confidentiality
 - Secure storage practices
-

13. BACKUP & RECOVERY

Database Backups

Recommended:

Daily backups.

File Backups

Recommended:

Cloud storage redundancy.

Recovery Procedures

The system shall support:

- Database restoration
 - File restoration
 - Disaster recovery processes
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14. SCALABILITY CONSIDERATIONS

Current Goals

Support:

- Small schools
 - Medium schools
 - Large secondary schools
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Future Growth

Potential support:

- Multi-school deployments
 - SaaS platforms
 - Additional enterprise modules
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Architectural Principles

The system shall avoid:

- Tight coupling
 - Hardcoded assumptions
 - Poor separation of concerns
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15. FUTURE TECHNICAL EXPANSION

Potential future additions include:

Parent Portals

Capabilities:

- Fee balances
 - Academic reports
 - Announcements
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SMS Services

Potential integrations:

- Africa's Talking
 - MTN APIs
 - Airtel APIs
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Enterprise Modules

Examples:

- Library systems
 - Inventory systems
 - Hostel management
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Analytics

Examples:

- Financial dashboards
 - Attendance insights
 - Enrollment statistics
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STRATEGIC CONCLUSIONS

The Natural Intellects School Management System shall prioritize:

- Simplicity
- Modularity
- Mobile responsiveness
- Data integrity
- Easy maintenance
- Cloud AI compatibility
- Future scalability

Version 1 shall follow:

One School

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One Database

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One Deployment

The architecture shall remain flexible enough to support future growth without introducing unnecessary complexity.

End of Document 06.